(MATH101, 102, 311, 312) Calculus and Analysis

Mathematical Concepts

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Contents

0 Notation and Conventions

General mathematical notation and writing follow *(MATH000) Mathematical Language*. The conventions below record only notation specific to calculus and analysis used in this book.

Single-Variable Calculus

The differential d is upright, and ε is the default small positive quantity in analysis.

Notation	Meaning
$\lim_{x\to a} f(x)$	limit as x approaches a
$\min A$	minimum of A
$\max A$	maximum of A
$\inf A$	infimum of A
$\sup A$	supremum of A
$f'(a)$	first derivative of f at a
$f^{(n)}(a)$	nth derivative of f at a
$\dfrac{\mathrm{d}y}{\mathrm{d}x}$	derivative of y with respect to x
$\displaystyle\int_a^b f(x)\,\mathrm{d}x$	definite integral over $[a,b]$
$\ln x,\ \log_e x$	logarithm to base e
$\lg x,\ \log_{10} x$	logarithm to base 10
$\operatorname{lb} x,\ \log_2 x$	logarithm to base 2
$\log_a x$	logarithm to base a
$\arcsin x$	principal inverse sine
$\arccos x$	principal inverse cosine
$\arctan x$	principal inverse tangent
$\operatorname{arcsec} x$	principal inverse secant
$\operatorname{arccsc} x$	principal inverse cosecant
$\operatorname{arccot} x$	principal inverse cotangent
$\Gamma(z)$	Gamma function
$B(x,y)$	Beta function
$\zeta(s)$	Riemann zeta function

Series and Convergence

Notation	Meaning
$\displaystyle\sum_{k=1}^{n} a_k$	finite sum
$\displaystyle\sum_{k=1}^{\infty} a_k$	infinite series
$\displaystyle\prod_{k=1}^{n} a_k$	finite product
$x_n \to x$	convergence of a sequence
$a_n \nearrow L$	monotone increasing convergence to L

Notation	Meaning
$a_n \searrow L$	monotone decreasing convergence to L
$\limsup a_n$	limit superior
$\liminf a_n$	limit inferior
$f_n \to f,\ f_n \xrightarrow{\text{ptw}} f$	pointwise convergence
$f_n \rightrightarrows f,\ f_n \xrightarrow{\text{unif}} f$	uniform convergence
$f_n \xrightarrow{\text{a.e.}} f$	convergence almost everywhere
$C^k(I)$	k times continuously differentiable functions on I
$C^\infty(I)$	smooth functions on I
$\\|f\\|_\infty$	supremum norm
$B_r(x)$	open ball of radius r centered at x
$\overline{B}_r(x)$	closed ball of radius r centered at x

Multivariable Calculus

Notation	Meaning
$\dfrac{\partial f}{\partial x_i},\ \partial_i f$	partial derivative with respect to x_i
$D\mathbf{F}(\mathbf{a})$	total derivative; Jacobian matrix in coordinates
$\operatorname{Jac}(\mathbf{F})$	Jacobian determinant
$\operatorname{Hess} f$	Hessian matrix of f
∇f	gradient of f
$\nabla \cdot \mathbf{F}$	divergence of $\mathbf{F}$
$\nabla \times \mathbf{F}$	curl of $\mathbf{F}$
Δf	Laplacian of f
$\iint_D f \ \mathrm{d}A$	double integral over D
$\iiint_\Omega f \ \mathrm{d}V$	triple integral over Ω
$\mathbf{r}(t)$	parametrized curve
$\mathrm{d}s$	element of arc length
$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$	line integral along C
$\iint_S \mathbf{F} \cdot \mathbf{n} \ \mathrm{d}S$	flux integral through S

1 (MATH101) Calculus I

2 (MATH102) Calculus II

3 (MATH311) Analysis I

4 (MATH312) Analysis II